Lab 3 Report
Projectile Motion

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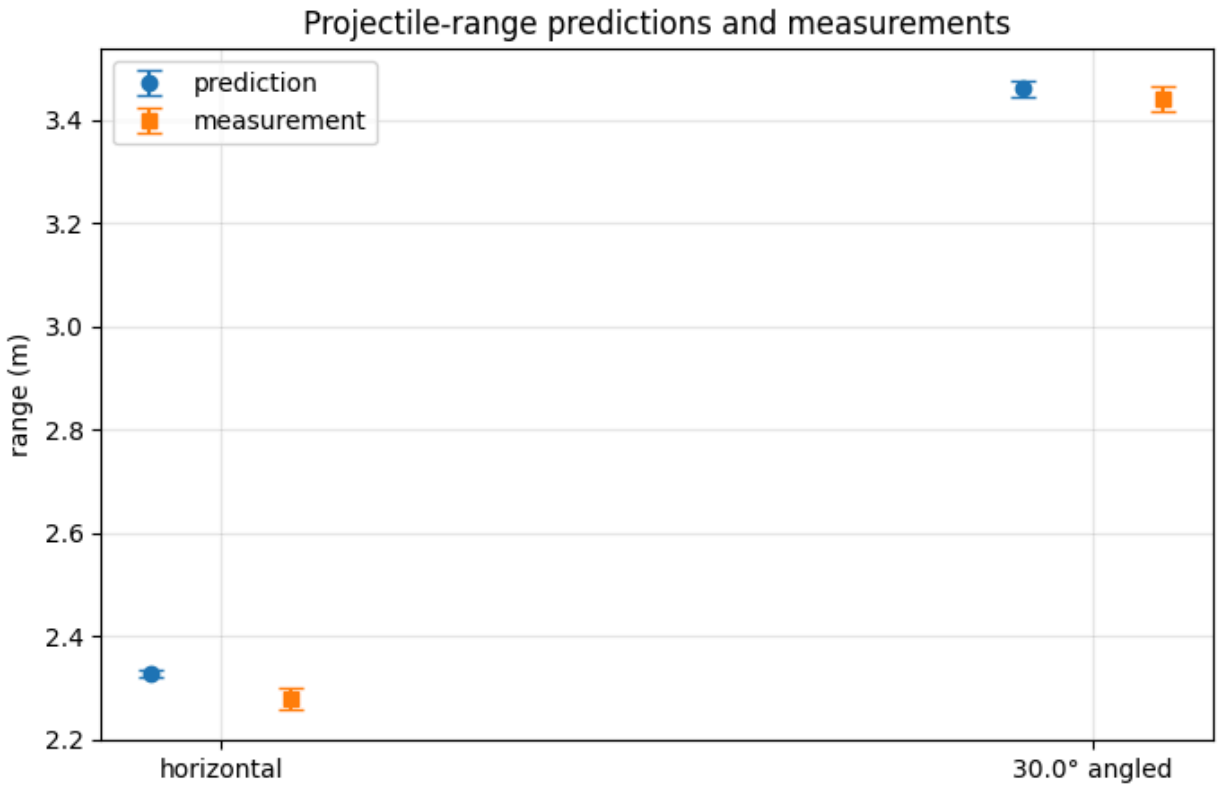
Mounting the launcher was done by clamping it down to the table to remain still, adjusting the launcher to our desired angle, then pushing the ball into our launcher with the provided tool until we heard 2 clicks putting it at a medium power level for our shots. We first tested the launcher without any measurements to see where it would land in relation to where we needed it to go, and then set the catcher and gathered the measurements we needed for the code. In some tests we had somebody in front of the launcher near the landing zone to stop the ball from rolling away, whoever was doing that job were provided safety goggles to protect themselves from the ball if it were to go awry.

The origin of the ball was 27.8cm above the table top. The origin of the meter stick is at the table top. Thus, for part 1 we subtracted 27.8 cm from each vertical meter stick measurement. We decided to use the same origin for the horizontal axis. To find the distance a meter stick was used to measure 200 Cm from the launcher. Then the location of impact was marked by the charcoal paper being pressed down by the ball. We then measured the length from the point of impact to the original 200cm reference point. These measurements gave us the length from the launcher to the platform.

We assigned uncertainties based on the selected tools of measurement, as well as what we were attempting to measure. More precisely, we used a minimum uncertainty for the measurements based on half the smallest unit possible for the instrument to measure and then added extra uncertainty based on how much approximation was required to measure to get the measurements, like how we ultimately used an uncertainty of +/- 1 because of how we seemed to be off of the exact middle of the ball within the launcher.

When measuring vertical launches a meter stick was placed upright on top of the table. Then using a phone the approximate height was measured. Using that measurement the distance from the table to the position of the ball was subtracted to gain a final measurement. For horizontal launches a piece of paper under a charcoal film was used to get imprints of the ball's landing point. Once these points had been gathered a ruler was used to measure the distance from the base of the platform to the imprint. This measurement was then added to the initial length measurement.

We practiced troubleshooting in how we continually checked the alignment of the launcher to ensure that it was flush with the table. Because of the continual misalignment due to recoil, we needed to scrap many previous attempts to make sure that we had proper measurements.



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index	configuration	predicted range (m)	prediction uncertainty (m)	measured range (m)	measurement uncertainty (m)	compatibility z
0	horizontal	2.326899224289698 4	0.007234947312327812	2.2800000000000002	0.020842264752180886	2.125764498505611
1	30.0° angled	3.4607474392348916	0.015563107256713057	3.4409	0.02552910104175234	0.6638179724378112

	trial	vertical rise (m)
0	1	1.323
1	2	1.317
2	3	1.320
3	4	1.322
4	5	1.321

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N = 5
mean = 1.3206 m
sample SD = 0.0023 m
standard error = 0.00103 m
uncertainty from meterstick resolution: 0.005 m
uncertainty from repeated trials: 0.00103 m
Adopted vertical-rise uncertainty: 0.005 m

Launch speed from vertical shots: 5.0874 ± 0.0097 m/s
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	trial	horizontal range (m)
0	1	2.226
1	2	2.257
2	3	2.259
3	4	2.329
4	5	2.329

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N = 5
mean = 2.28 m
sample SD = 0.0466 m
standard error = 0.0208 m
Measured horizontal range: 2.2800 ± 0.0208 m
Prediction-measurement compatibility: z = 2.13
Launch speed from horizontal shots: 4.9848 ± 0.0472 m/s

Predicted horizontal flight time: 0.4574 s
Predicted horizontal range: 2.3269 ± 0.0072 m
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	trial	angled range (m)
0	1	3.3830
1	2	3.4290
2	3	3.4725
3	4	3.5225
4	5	3.3975

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N = 5
mean = 3.4409 m
sample SD = 0.0571 m
standard error = 0.0255 m
Measured angled range: 3.4409 ± 0.0255 m
Prediction-measurement compatibility: z = 0.66

Predicted range at 30.0°: 3.4607 ± 0.0156 m
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The predicted and measured ranges agreed with one another. Horizontal shots were predicted to be 2.32 Meters and measured to be 2.28 Meters. Angled horizontal shots were predicted to be 3.46 Meters and measured to be 3.44 Meters. Our Z values were within the given range of 2 except for the horizontal method which was slightly above at 2.28. The launch speed for our vertical test was slightly faster than the horizontal method which we concluded could have been due to issues with the launcher and its recoil and our troubles in getting the launcher to be exactly 0 degrees while setting it at 90 degrees was much easier and precise. As a group, we believe that angle alignment and recoil from repeated launches were the most likely causes for the differences and increased Z value for the horizontal test when compared to the other tests. To get more precise results, we would use a more precise way of checking the angle of the launcher as opposed to the plumb bob used to measure the angle and find a more powerful clamp to keep the launcher in place and reduce recoil from repeated launches in such a concentrated amount of time.